

Overlapping Generations in Lean: The Life-Cycle Household

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Abstract

This is the record of a programme to formalise overlapping-generations economies in Lean 4, and of its first stage: the life-cycle household. A household that lives J periods against uninsurable earnings risk and a borrowing constraint solves a finite backward induction, and everything proved for the infinite-horizon Bellman operator with an arbitrary continuation applies stage by stage with no fixed point. Beyond that we prove the finite-horizon sandwich: with k periods still to come, consumption lies between $\kappa_k m$ and $\kappa_k(m + H_k(z))$, where $1/\kappa_k = \sum_{i \leq k} (b/R)^i$ is the consumption share of a household with no future income and H_k is the expected present value of the next k incomes; the propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - b/R$. We then add retirement and a pension. With a lump-sum pension the retired phase is an income-fluctuation problem in its own right and everything survives. With a means-tested pension we settle a claim often made informally: a means test does make the policy non-monotone, but not for the reason usually given. Optimal saving rises with cash on hand whatever the continuation, by Topkis, so a value function made non-concave by the test cannot by itself reverse it; what the test reverses is consumption, which jumps down at the wealth where the household gives the pension up, and cash on hand itself, once the taper is steeper than the gross return. The programme's target is uniqueness of the stationary equilibrium at relative risk aversion above one, which the infinite horizon could not close: in a life cycle every comparative static is a finite induction, wealth is bounded, and the age-one distribution is common to every rate.

JEL: C63, D52, E21. *Keywords:* overlapping generations, life cycle, Bewley models, formal verification, Lean.

1 Introduction

The companion write-up in this repository proved that the stationary equilibrium of Aiyagari (1994) is unique at his logarithmic calibration, and located exactly what stands in the way

*The Lean development described here is at <https://github.com/LeanEconomics/LeanEconomics>. It was written mostly by Claude (Anthropic), an AI coding model, working under the direction of Robert Kirkby (Victoria University of Wellington); every result reported was checked by the Lean kernel. This is the companion to the write-up on Aiyagari (1994) in the same repository. Comments welcome.

for relative risk aversion $\mu \in \{3, 5\}$. Every obstacle it met traces to the fixed point of the infinite-horizon problem: a condition on the response of consumption to the interest rate fails in a tail of wealth that the stationary distribution reaches; an induction that would prove the response bound has a step multiplier $1/\bar{p} > 1$, so it cannot close as a supremum argument; and the only floor on capital supply comes from an ergodic identity that loses a factor $\|u'\|^2/\text{Var}$. Overlapping generations remove all three at once. A household that lives J periods solves a finite backward induction from an explicit terminal condition, so a multiplier $1/\bar{p}$ per step compounds to $\bar{p}^{-J}$, which at Aiyagari's numbers is about 1.03. Wealth is bounded by J periods of accumulation, so any condition needs checking on a known bounded region. And the distribution of the youngest cohort is common to every interest rate, so an ordering of policies pushes forward to an ordering of the whole age profile of distributions by induction on age, with no Doeblin atom and no uniqueness of a stationary distribution.

The target of the programme is uniqueness of the stationary equilibrium of a stochastic life-cycle economy in the manner of Huggett (1996) at $\mu \in \{3, 5\}$. Section 2 records the first stage, the life-cycle household, which is formalised. Section 4 records the rest of the programme, the classical results of Diamond (1965) included, and what each stage needs. As in the companion, every result stated with a formal name has been checked by the Lean kernel against Mathlib; the development is at <https://github.com/LeanEconomics/LeanEconomics>, directory `LeanEconomics/OLG`.

2 The life-cycle household

2.1 Stages

The household of the companion write-up is kept: earnings $y(z)$ follow a finite Markov chain π , the gross return is $R = 1 + r$, cash on hand is $m = Ra + y(z)$, saving $a' \in [0, \bar{a}]$ is bounded below by the borrowing limit and above by an artefactual cap, utility is CRRA with $u'(c) = c^{-\mu}$ and $\beta \in (0, 1)$. What changes is the horizon: the household lives J periods and leaves nothing behind. Count periods from the end. With k periods still to come after today, the continuation value is the k -th Bellman iterate from zero,

$$V_k = T^k 0, \quad (Tv)(a, z) = \max_{a'} \{u(m - a') + \beta \sum_{z'} \pi(z, z') v(a', z')\},$$

and the household with k periods to come saves $g_k = \text{policyOf } V_k$ and consumes $c_k = m - g_k$ (`stageValue`, `stagePolicy`, `stageConsumption`). Age j of a J -period life is stage $k = J - 1 - j$. In the last period the household saves nothing and consumes its cash on hand (`stagePolicy_zero`, `stageConsumption_zero`).

This is the finite-horizon problem with a flat age profile of earnings: the same income process at every age. An age profile, retirement and survival risk need income to be a parameter of the Bellman step rather than a field of the structure, and are deferred to the next stage of the programme.

2.2 What transfers without a fixed point

The companion development proved its household results for the Bellman operator applied to an arbitrary continuation v , and only then passed to the fixed point. Here there is no fixed point to pass to, and each result is read at $v = V_k$:

- consumption is positive at every stage when marginal utility is unbounded at zero (`stageConsumption_pos`);
- the stage value has concave slices, so the optimal action is unique and both saving and consumption rise with wealth (`stagePolicy_mono`, `stageConsumption_mono`);
- Carroll–Kimball: consumption is concave in wealth at every stage (`concaveOn_stageConsumption`);
- the Euler inequality under the cap, $\beta R \mathbb{E}[u'(c_k(a', z')) \mid z] \leq u'(c_{k+1}(a, z))$, at every state (`stage_euler_le`), and the Euler inequality above the floor, the reverse, at every state where the household saves (`stage_euler_ge`).

2.3 The finite-horizon sandwich

Let $\mathfrak{p} = (\beta R)^{1/\mu}$ and define the finite-horizon minimal propensities to consume by

$$\kappa_0 = 1, \quad \kappa_{k+1} = \frac{\kappa_k R}{\mathfrak{p} + \kappa_k R}, \quad \text{so that} \quad \frac{1}{\kappa_k} = \sum_{i=0}^k \left(\frac{\mathfrak{p}}{R}\right)^i$$

(`stageMPC`): the share of cash on hand consumed by a household with k periods to come and no future income. Let $H_0 = 0$ and $RH_{k+1}(z) = \sum_{z'} \pi(z, z')(y(z') + H_k(z'))$, the expected present value of the next k incomes (`stageHumanWealth`).

Theorem 2.1 (The finite-horizon sandwich). *Assume CRRA utility with marginal utility unbounded at zero, and that the cap is slack at every stage. Then for every k , every z and every $a \in [0, \bar{a}]$,*

$$\kappa_k m \leq c_k(a, z) \leq \kappa_k (m + H_k(z)).$$

Formal statement: `IncomeFluctuation.stageMPC_mul_le_stageConsumption`,
`IncomeFluctuation.stageConsumption_le_stageHumanWealth`.

Each bound is one Euler step, applied k times. The lower bound: if consumption against v is at least κm , the Euler inequality above the floor at a state where the household saves gives $c^{-\mu} \leq \beta R \mathbb{E}[(\kappa R a')^{-\mu}]$ with $a' = m - c$, hence $\kappa R(m - c) \leq \mathfrak{p}c$ and $c \geq \kappa R(\mathfrak{p} + \kappa R)^{-1}m$; at a corner $c = m$ (`crra_stageMPC_step`). The upper bound: if consumption against v is at most $\kappa(m + H(z))$, the Euler inequality under the cap and Jensen for the negative power give $c^{-\mu} \geq \beta R (\mathbb{E}[c'])^{-\mu}$ with $\mathbb{E}[c'] \leq \kappa R(a' + H'(z))$, hence $c \leq \kappa R(\mathfrak{p} + \kappa R)^{-1}(m + H'(z))$ (`crra_stageUpper_step`). The recursion for κ is the same on both sides, which is why the sandwich is exact in the homogeneous case.

The propensities decrease in the horizon towards the infinite-horizon $\kappa = 1 - \mathfrak{p}/R$ of the companion write-up: $\kappa \leq \kappa_k$ for every k (`minMPC_le_stageMPC`), so a household with less life left consumes a larger share of its cash, and the infinite-horizon bounds are the limiting case. A first consequence for the programme is an explicit bound on saving, $g_k(a, z) \leq (1 - \kappa_k)(Ra + y_{\max})$ (`stagePolicy_le`), which iterated from zero wealth at the start of life bounds the wealth of every cohort.

3 Retirement, a pension, and the means test

3.1 Retirement with a lump-sum pension

The household works until age J_r and is retired after it, receiving a pension in place of its labour earnings. With a *lump-sum* pension p the retired household faces the same problem as the working one with income replaced by the constant p , so the retired phase is itself an income-fluctuation problem (`withPension`) and everything in Section 2 applies to it unchanged. The life-cycle value is assembled by iterating the working Bellman operator on the retired value: `lifeValue( $n_r, n_w$ )` is the value of a household with n_w working and n_r retired periods still to come (`lifeValue`), so a household of age $j < J_r$ in a life of J periods sits at $n_w = J_r - j$ and $n_r = J - J_r$. Saving and consumption rise with wealth at every age (`lifePolicy_mono`, `lifeConsumption_mono`).

Two facts about the pension itself are worth recording. A retiree's human wealth is the same in every earnings state, because the pension does not depend on the earnings shock (`stageHumanWealth_withPension_const`), and it never exceeds the perpetuity value of the pension, $H_k \leq p/r$ (`stageHumanWealth_withPension_le`). With Theorem 2.1 the retiree's consumption therefore satisfies

$$\kappa_k m \leq c_k(a) \leq \kappa_k \left(m + \frac{p}{r} \right)$$

at every retired age. And cash on hand is strictly increasing in assets (`withPension_resources_strictMono`). Both of these fail once the pension is means-tested.

3.2 Means testing

A pension is means-tested when the amount paid falls with the recipient's assets. We write the schedule as a function $p(\cdot)$ of assets and cash on hand as $m(a) = Ra + p(a)$ (`cash`), and consider three schedules (`lumpSum`, `tapered`, `cliff`): a lump sum; a taper, withdrawn at rate τ per unit of assets above a threshold and never negative; and a cliff, withdrawn entirely above the threshold.

It is often argued that a means test makes the household's policy function non-monotone, and the reason usually given is that the test makes the value function non-concave. The argument as usually given is wrong, and the conclusion is nevertheless true, for a different reason. Three results separate the two.

Theorem 3.1 (Monotonicity in cash on hand is never lost). *Let period utility be strictly concave on the positive half-line and let W be any continuation value. If $m_1 < m_2$ and x_i is optimal at m_i , then $x_1 \leq x_2$.*

Formal statement: *OLG.isOptimalSaving_mono.*

The proof is Topkis. Adding the two optimality inequalities gives $u(m_1 - x_1) + u(m_2 - x_2) \geq u(m_1 - x_2) + u(m_2 - x_1)$; if $x_2 < x_1$ the two right-hand consumptions are strict convex combinations of the two left-hand ones with the same sum, so strict concavity makes the reverse inequality strict, a contradiction. The continuation enters only through terms that cancel. So a non-concave value function — which a means test certainly produces — cannot by itself make saving fall with cash on hand, whatever the shape of W .

What a non-concave continuation does produce is a jump, and the jump falls on consumption.

Theorem 3.2 (Consumption falls when wealth rises). *There are a continuation value arising from a means-tested pension and two levels of cash on hand, the second larger, at which optimal consumption is strictly smaller. With log utility, a unit gross return, and a pension of one withdrawn entirely above assets of one: at $m = 6$ the household saves exactly 1 and consumes 5; at $m = 8$ it saves 4 and consumes 4.*

Formal statement: *OLG.exists_consumption_decrease.*

At $m = 6$ the household keeps the pension, saving the most it can without losing it; at $m = 8$ the pension is no longer worth keeping, so the household gives it up and saves the proceeds. Saving rises from 1 to 4, as Theorem 3.1 says it must, and since it rises by more than wealth does, consumption falls. Both optima are exact: the objective is log of a quadratic on each side of the threshold, and the comparison is checked by the kernel.

The third result is the genuine non-monotonicity of the saving policy, and it comes from the budget rather than from the value function. A retiree's pension is assessed each period on the assets then held, so the schedule enters cash on hand directly.

Theorem 3.3 (A gentle taper is harmless; a cliff is not). *If $0 \leq \tau < R$ then $a \mapsto Ra + p(a)$ is strictly increasing under the taper (*cash_strictMono_tapered*), and so, by Theorem 3.1, saving is monotone in assets (*isOptimalSaving_mono_assets*). Under a cliff it is not: if $a \leq \bar{a} < b$ and $R(b - a) < p$ then cash on hand is strictly smaller at b (*cash_lt_cash_of_cliff*), and there are two asset levels, the second larger, at which the household saves strictly less: with a concave continuation, at assets 1 it saves $1/2$ and at assets $3/2$ only $1/4$.*

Formal statement: *OLG.exists_saving_decrease_in_assets.*

The continuation in the second half of Theorem 3.3 is $\log(x + 1)$, which is concave: nothing is hidden in it. The non-monotonicity is entirely the budget's, and the threshold is sharp, τ against R . A taper gentler than the gross return leaves the household's saving monotone in assets no matter how badly the test distorts the value function; a taper steeper than the gross return, or a cliff, makes a richer household poorer in the only sense that matters for today's decision.

3.3 Which policy, and in which variable

The two non-monotonicity results are about different objects, and it is worth separating them in the household's own state space, assets a and the Markov income shock z .

Next period's assets are monotone. For each z , the optimal a' is nondecreasing in a whatever the continuation value, with no concavity assumed of it:

Theorem 3.4 (a' rises with a , at each z). *Let $a < a'$ both lie in the asset region. Then for every continuation v and every z , $g_v(a, z) \leq g_v(a', z)$.*

Formal statement: *IncomeFluctuation.policyOf_mono_of_lt.*

This is Theorem 3.1 read in the state space: cash on hand rises with assets, so optimal saving does. The repository already had this statement (`policyOf_mono`) but with concave slices of the continuation as a hypothesis, which is exactly what a means test destroys; the Topkis argument removes the hypothesis, at the cost of requiring the two asset levels to be distinct. The continuation cancels between the two optimality inequalities and only strict concavity of period utility survives.

Consumption is not monotone, and the contrast is sharp rather than an artefact of proof technique. Consumption is $m(a) - g_v(a, z)$, and both terms rise with a ; when the continuation is non-concave the second can rise faster, which is exactly Theorem 3.2. So monotone consumption genuinely requires concavity of the continuation, while monotone saving does not.

Two things follow for the means test. Where the pension enters only next period's problem, so that today's cash on hand is unaffected, Theorem 3.4 applies verbatim: next period's assets still rise with this period's, and only consumption misbehaves. Where the test is assessed on the assets currently held, the hypothesis of Theorem 3.4 is the one that fails: cash on hand is no longer increasing in a , and with it the policy. That is the whole of Theorem 3.3, and the dividing line is τ against R .

3.4 At realistic parameters

The three theorems predict exactly which of the two channels operates for which schedule, and a calibrated retiree confirms it. We solve a deterministic retiree of 25 years by backward induction with global search over a grid of 6001 asset levels — the value function is not concave, so a first-order condition cannot be used — at $r = 4\%$, $\beta = 0.96$, relative risk aversion 3, a full pension of 0.40 of mean earnings and a free area of 3 mean-earnings units (`numerics/meanstest.m`). The taper of 0.078 per unit of assets is Australia's asset test, three dollars a fortnight per thousand dollars.

Pension schedule	Cash on hand	Saving	Consumption
Lump sum	increasing	increasing	increasing
Taper 0.078 (Australia)	increasing	increasing	falls by 0.025–0.046
Taper 0.50	increasing	increasing	falls by 0.13–0.24
Taper 1.20 (above R)	falls	falls by 0.004	falls by 0.55–0.70
Cliff	falls by 0.40	falls by 0.32–0.37	falls by 0.55–0.70

Every entry is what the theorems require. Saving falls with assets in exactly the two rows where cash on hand does, which are exactly the two rows whose taper exceeds the gross return, as Theorem 3.3 says; in the other three it rises, as Theorem 3.1 says it must. Consumption falls in every row with a test of any kind, as Theorem 3.2 says it can, and the drop sits exactly at the free area and grows with age, from 0.025 of mean earnings at the start of retirement to 0.046 twenty years in under the Australian taper. The mechanism is the one the witness isolates: just above the free area the effective return to saving is only $R - \tau$ until the pension is exhausted, so the household jumps its saving toward the far side of the taper range and cuts consumption to pay for it.

The effect is not confined to retirees. A worker in the last year before retirement, facing the Australian taper for the whole of retirement, has a monotone saving policy and a consumption policy that falls by 0.025 of mean earnings at the same threshold: the non-concavity is inherited through the continuation value, and it is again consumption that carries it.

4 The programme

The stages, in the order we intend to take them. The first is done (Section 2).

Equilibrium. A stationary overlapping-generations equilibrium: the distribution μ_1 of the youngest cohort is given (zero wealth, earnings drawn from the invariant law), μ_{j+1} is the pushforward of μ_j under the age- j policy and the earnings chain, capital supply is $K(r) = \sum_j \omega_j \int a d\mu_j$ with cohort weights ω_j , labour is exogenous, and the firm is Aiyagari's. Targets: continuity of K in the rate, a finite composition of continuous pushforwards; existence by the intermediate value theorem against the firm's demand, with the floor at the top of the interval from explicit accumulation over the life rather than from an ergodic identity; Light's Theorem 2 by induction on age, an ordering of policies at every age giving first-order stochastic dominance of every μ_j and so of K ; and single crossing against a decreasing demand.

Theorem 1 by backward induction. The research core: saving rises with the rate at every age, for $\mu \in \{3, 5\}$. The reduction of the companion write-up applies stage by stage, Theorem 1 at stage $k + 1$ following from the slack elasticity condition on the stage- k consumption functions at the pivot; the transfer lemma propagates an affine bound on the response of consumption to the rate backward from the terminal condition $c_0 = m$, where the response is exactly $\Delta r \cdot a$; and the slack condition at each stage is a finite check on the constants so obtained. The obstruction of the infinite horizon, a step multiplier of at least $1/p$, does not bite in J steps. The first session is numerical: a life-cycle economy with $J = 60$, Aiyagari's earnings chain and $\mu \in \{3, 5\}$, checking the slack condition at every age and wealth and running the recursion for the constants. If it holds the argument is formalised; if it fails only at moderate ages and wealth we look for the age-dependent refinement; if it fails badly, log utility is the fallback, where Theorem 1 at each stage follows from the rescaling proof of the companion, which is already an induction on Bellman iterates.

Diamond (1965) and Samuelson. The two-period economy without risk: a unique steady state and monotone global convergence for logarithmic utility and Cobb–Douglas production, existence under Inada conditions and a multiplicity witness in general, dynamic inefficiency of a steady state with $r < n$ by an explicit Pareto-improving transfer, the golden rule, and the autarkic and monetary steady states of the pure-exchange economy. These are standalone results and will be the classical sections of this write-up.

Extensions. An age profile of earnings, retirement with a pension and a government budget, survival risk with accidental bequests as a second fixed point, population growth, and Lasry–Lions with age as time, whose finite-horizon core is already formalised and whose monotonicity is again the elasticity condition.

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A Index of formal results

All in the namespace `LeanEconomics.IncomeFluctuation`, directory `LeanEconomics/OLG`. Every result depends only on the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and none uses `sorry`.

Result	Lean name	Module
Stage value, policy, consumption	<code>stageValue,</code> <code>stageConsumption</code>	<code>stagePolicy,</code> <code>OLG/LifeCycle</code>
Last period: no saving	<code>stagePolicy_zero</code>	<code>OLG/LifeCycle</code>
Last period: consume cash on hand	<code>stageConsumption_zero</code>	<code>OLG/LifeCycle</code>
Positive consumption at every stage	<code>stageConsumption_pos</code>	<code>OLG/LifeCycle</code>
Saving and consumption rise with wealth	<code>stagePolicy_</code> <code>stageConsumption_mono</code>	<code>mono,</code> <code>OLG/LifeCycle</code>
Carroll–Kimball at every stage	<code>concaveOn_stageConsumption</code>	<code>OLG/LifeCycle</code>
Euler under the cap, by stage	<code>stage_euler_le</code>	<code>OLG/LifeCycle</code>
Euler above the floor, by stage	<code>stage_euler_ge</code>	<code>OLG/LifeCycle</code>
Finite-horizon propensities, human wealth	<code>stageMPC, stageHumanWealth</code>	<code>OLG/LifeCycle</code>
Propensities in $(0, 1]$, above κ	<code>stageMPC_pos,</code> <code>stageMPC_le_one,</code> <code>minMPC_le_stageMPC</code>	<code>OLG/LifeCycle</code>
Lower Euler step	<code>crra_stageMPC_step</code>	<code>OLG/LifeCycle</code>
Upper Euler step	<code>crra_stageUpper_step</code>	<code>OLG/LifeCycle</code>

Result	Lean name	Module
Theorem 2.1, lower	<code>stageMPC_mul_le_stageConsumption</code>	OLG/LifeCycle
Theorem 2.1, upper	<code>stageConsumption_</code> <code>stageHumanWealth</code> <code>le_</code>	OLG/LifeCycle
One-step saving bound	<code>stagePolicy_le</code>	OLG/LifeCycle
Retired household (lump-sum pension)	<code>withPension</code>	OLG/Retirement
Cash on hand rises with assets	<code>withPension_resources_strictMono</code>	OLG/Retirement
Human wealth is state-independent	<code>stageHumanWealth_</code> <code>withPension_</code> <code>const</code>	OLG/Retirement
Perpetuity bound $H_k \leq p/r$	<code>stageHumanWealth_withPension_le</code>	OLG/Retirement
Life-cycle value with retirement	<code>lifeValue</code>	OLG/Retirement
Saving, consumption rise with wealth	<code>lifePolicy_mono</code> , <code>lifeConsumption_</code> <code>mono</code>	OLG/Retirement
Pension schedules	<code>lumpSum</code> , <code>tapered</code> , <code>cliff</code>	OLG/MeansTest
Cash on hand	<code>cash</code>	OLG/MeansTest
Theorem 3.1	<code>isOptimalSaving_mono</code>	OLG/MeansTest
Monotone in assets when cash is	<code>isOptimalSaving_mono_assets</code>	OLG/MeansTest
Gentle taper keeps cash rising	<code>cash_strictMono_tapered</code>	OLG/MeansTest
A cliff reverses cash on hand	<code>cash_lt_cash_of_cliff</code>	OLG/MeansTest
Theorem 3.2	<code>exists_consumption_decrease</code>	OLG/MeansTest
Theorem 3.3, cliff	<code>exists_saving_decrease_in_assets</code>	OLG/MeansTest
Theorem 3.4	<code>policyOf_mono_of_lt</code>	OLG/MeansTest